

Ioannis Petrousov

SYSTEMS & AI ENGINEER

Zuid Holland, Netherlands

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🎓 ResearchGate | 🤖 HuggingFace

Skills

Programming	Python, Bash, CMake, Lua, Typescript, Go
Infra & Cloud	AWS, Kubernetes, EKS, ECS, Docker, Terraform, Ansible, ArgoCD, GitHub Actions
AI & ML	PyTorch, HuggingFace, smolagents, Ollama, ChromaDB (Vector DB), scikit-learn, Pandas
Web	FastAPI, Streamlit, HTML, CSS
Hardware & EDA	Logisim, Modelsim, Quartus, VHDL
Operating systems	Linux & Unix based OSs, Manjaro
Languages	Greek, English, Russian, Dutch (A2)

Education

UniPi (University of Piraeus) & NCSR (National Centre for Scientific Research)

Athens, Greece

Demokritos

M.S. IN ARTIFICIAL INTELLIGENCE

2025-today

- **Thesis (Ongoing):** Deep Reinforcement Learning with Large Language Model Reflective Text Embeddings.
- Intense 18-months program on AI.
- Two times finalist at ACE AUEB startup competition: Prototyped and pitched a data extraction engine from unstructured PDF documents.

UWM (University of Western Macedonia)

Kozani, Greece

M.S. IN INFORMATICS AND TELECOMMUNICATIONS ENGINEERING

2009-2017

- **Thesis Published):** Automating the generation of hardware accelerators from custom arithmetic functions.
- **Publications:** Authored and co-authored 9 peer-reviewed publications in IEEE societies focusing on **EDA (Electronic Design Automation)** tools, hardware accelerator generation, and IP block optimization.

Experience

VodafoneZiggo

Utrecht, Netherlands - Hybrid

SENIOR FREELANCE CLOUD DEVELOPER

Jun 2024 - Jan 2025

- Spearheaded the migration of middleware from on-prem to AWS EKS, designing architecture for massive scalability and cost optimization.
- Automated complex repetitive tasks and infrastructure logic using Python Lambdas (DNS, certificates, logging).
- Engineered self-service infrastructure modules in Terraform and highly concurrent CI/CD pipelines for package releases.
- Orchestrated stateful deployments and distributed computing across 61 AWS accounts.

101 Ways (Client: Recharge)

Amsterdam, Netherlands -

Hybrid

SENIOR FREELANCE DEVELOPER

Nov 2023 - Jan 2024

- Consulted on architecture and engineered the V2 platform rollout using AWS EKS and CDKTF (TypeScript).
- Built dynamic automated CI/CD pipelines with GitHub Actions to spin up ephemeral environments and manage container storage.

DPG Media

FREELANCE DEVOPS ENGINEER

Amsterdam, Netherlands -

Hybrid

Jun 2023 - Oct 2023

- Architected the production AWS ECS landing zone and developed modular Terraform code to provision segregated, scalable environments.
- Engineered custom Python deployment pipelines and automated resource cost-controls via AWS Lambdas.

ZEN Software (Client: Maxeda)

FREELANCE CLOUD ENGINEER

Alkmaar, Netherlands - Hybrid

Mar 2022 - Dec 2022

- Architected an automated CI/CD lifecycle via BitBucket Cloud, engineering custom self-hosted runners integrated with BuildKit for caching.
- Developed serverless Python Lambda logic to dynamically toggle security challenges and automated infrastructure with Terraform.

Hubs - a Protolabs company

DEVOPS ENGINEER

Amsterdam, Netherlands

Jan 2020 - Mar 2022

- Developed the first Public ML pricing API using AWS API Gateway and Lambdas.
- Migrated CI pipelines for ML projects from Jenkins to GitHub Actions, performing optimizations with parallelization and smart caching.
- Provisioned self-hosted Kubernetes clusters on AWS and executed zero-downtime migrations.

Mirabeau - a Cognizant Digital Business (Clients: Karwei, Arcadia, Maxeda DIY)

SENIOR CLOUD SOLUTION ENGINEER

Amsterdam, Netherlands

Mar 2019 - Dec 2020

- Consulted multiple dev teams on infrastructure architecture, CI/CD, and executed PoC migrations of legacy systems to AWS.
- Migrated self-hosted HTTP Apache edge servers into scalable microservices and acted as the go-to person for edge routing.
- Wrote infrastructure-as-code in Terraform and Ansible, and developed custom Python Lambdas for system glue logic.

PVH Corp.

DEVOPS SPECIALIST

Amsterdam, Netherlands

May 2018 - Feb. 2019

- Built the centralized API hub for all mobile apps using Kong API Gateway.
- Reverse-engineered proprietary Kong plugins and developed custom Lua APIs for middleware systems.

Betiator gaming technologies

SYSTEM ADMINISTRATOR

Athens, Greece

Aug. 2015 - Oct. 2016

- Implemented a HA platform based on Docker, Mesos, Marathon and Zookeeper, later migrated to DC/OS.
- Developed an automation framework for infrastructure autoscaling, recovery, and reconfiguration.

Certifications & Awards

- 2023 **AWS Solutions Architect - Associate**, XV6M9JC2EBF41Q32
- 2022 **Certified Kubernetes Application Developer (CKAD)**, LF-p2hxp1v6o8
- 2020 **AWS Certified DevOps Engineer - Professional**, WX0CEL01HJBEQJ97
- 2020 **AWS Certified Developer - Associate**, RZB70KQ1PMRQ1MS2
- 2019 **AWS Certified Solutions Architect - Associate**, XV6M9JC2EBF41Q32
- 2014 **Rewarding scholarship**, University of Western Macedonia
- 2013 **Rewarding scholarship**, University of Western Macedonia

Greece

Greece

Projects & Interests

Self-Hosted Homelab: Architecting local-only smart home automations and exploring open-source infrastructure.